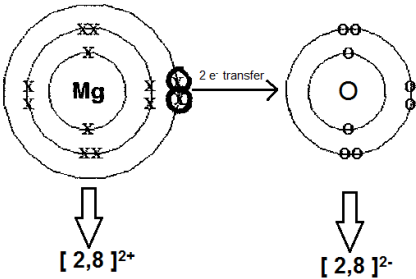
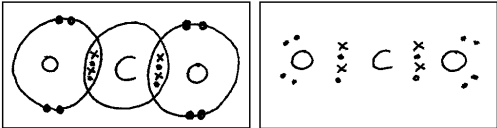


Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			 both outer shell electrons of the magnesium shown going to the outer shell of the oxygen (1) electron configuration and charge of magnesium ion (1) electron configuration and charge of oxide ion (1) award (1) for charges of both the magnesium and oxide ions without electron configurations / both configurations of the magnesium and oxide ions without charges accept outer shell electron diagrams only	1	2		3		
	(b)			the <u>charges</u> of the ions in <u>magnesium oxide</u> are <u>greater</u> (than the charges of the ions in sodium chloride) (1) this results in <u>greater attraction</u> / <u>stronger bonds</u> needing <u>more energy</u> to break down the structure (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			 diagram with two shared pairs of electrons in each overlap and complete octets (2) award (1) for overlapping both oxygen outer shells with carbon outer shell with one shared pair in each overlap		2		2		
				Question 2 total	3	4	0	7	0	0